

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Smart Study Generator Agent

Domain of Project – **[Education]**

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Problem Statement -

Problem Statement : Smart Study Generator Agent (StudyGenie)

The Challenge

Students often struggle with **planning their studies, finding reliable learning resources, and managing exam preparation**. Study materials are scattered across different platforms, leading to information overload and inefficient learning. Without **personalized guidance**, students find it difficult to stay organized and achieve their learning goals

The Objective

Build an **AI-powered Smart Study Generator Agent** that provides **personalized study plans and intelligent learning support**. The solution should:

- **Personalized Study Plans** – Create schedules based on subjects, goals, and available time.
- **RAG-Based Content Retrieval** – Fetch and summarize relevant information from trusted study materials.
- **Study Material Generation** – Generate notes, summaries, flashcards, and quizzes.
- **Progress Tracking** – Monitor learning progress and provide feedback.
- **Multi-Agent System**
 - **Study Planner Agent**
 - **Content Retrieval Agent**
 - **Quiz & Assessment Agent**
 - **Learning Assistant Agent**
- **Learning Dashboard** – Display study progress, completed topics, and quiz performance.

Proposed Solution -

The **Smart Study Generator Agent** is an AI-powered learning assistant that creates personalized study plans based on a student's learning goals, available study time, subject difficulty, and exam schedule. It generates structured daily and weekly study plans, recommends relevant learning resources, tracks progress, and adapts the schedule according to the user's performance. The system also provides reminders, revision plans, quizzes, and motivational insights to improve consistency and learning outcomes. By leveraging AI, the solution helps students study more efficiently, reduce time spent on planning, and focus on achieving their academic goals.

Key Features:

- Personalized study plan generation
- AI-based subject prioritization
- Daily and weekly schedules
- Progress tracking and performance analysis
- Automatic revision planning
- Quiz and practice question recommendations
- Smart reminders and notifications
- Adaptive learning based on user performance
- Resource recommendations (videos, notes, articles)
- Dashboard for monitoring study goals and achievements

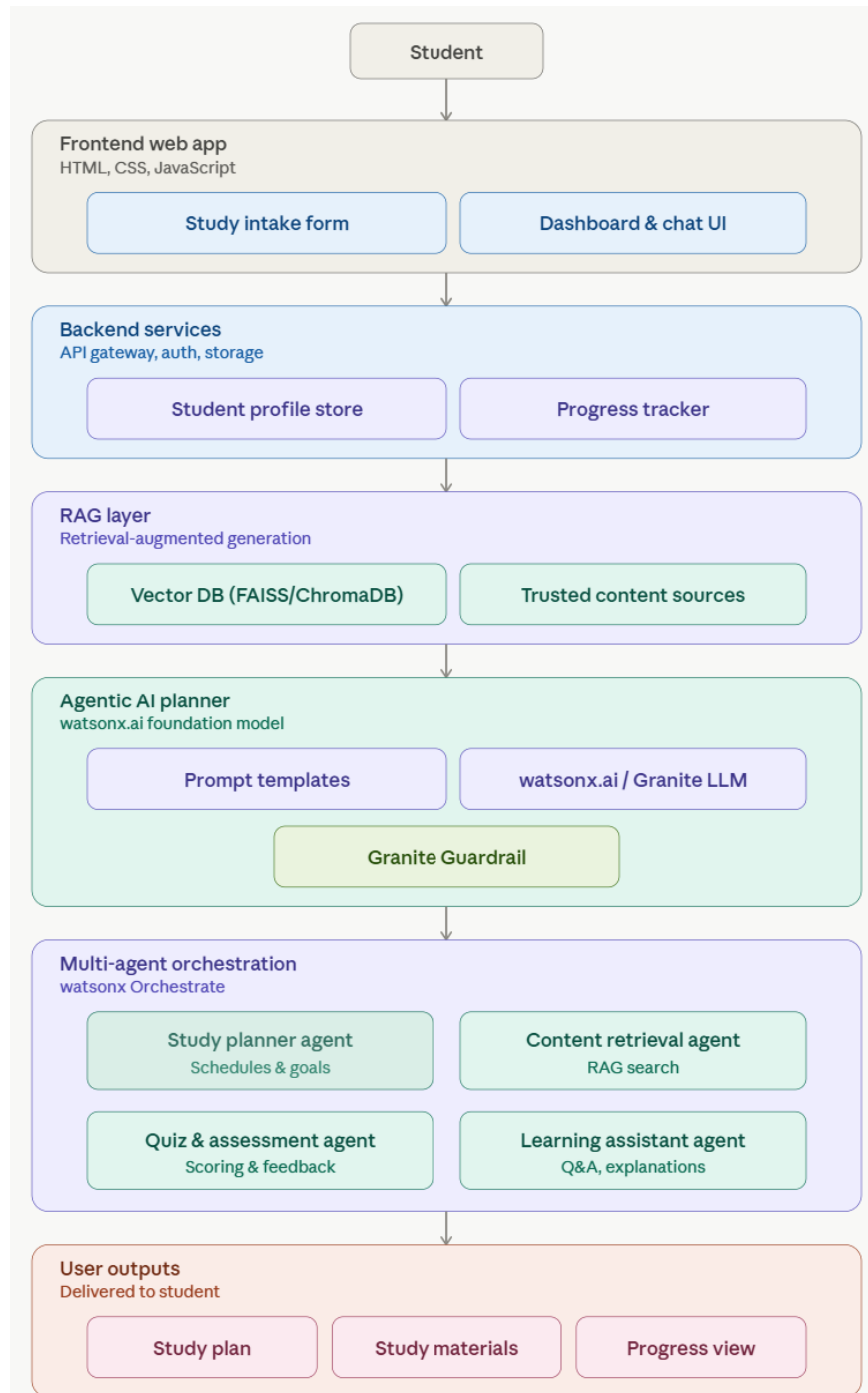
Technology Used

- **Watsonx Orchestrate platform** - IBM's AI-powered automation platform that orchestrates intelligent agents and workflows to automate tasks, streamline processes, and enhance productivity.
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization.
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Langflow component Used - Smart Study Generator Agent

- 1) **Chat Input** – Interface where students enter their queries, learning goals, topics, or upload study materials (text/PDF/image).
- 2) **IBM watsonx AI Agent** – Processes user input using IBM Granite models to generate personalized study plans, summaries, quizzes, and explanations.
- 3) **IBM Granite Model – Granite-4.0-8B-Instruct** → Fast and efficient for real-time study assistance, concept explanation, and content generation.
- 4) **Chat Output** – Displays AI-generated study plans, notes, summaries, practice questions, and personalized learning recommendations.
- 5) **File Component** – Stores and retrieves study materials, notes, PDFs, and reference documents for AI-powered analysis using RAG.
- 6) **Prompt Component** – Defines system instructions and learning workflows, ensuring accurate, structured, and context-aware educational responses.

Architecture blueprint



Role of Agentic AI in the solution

Role of Agentic AI in Nutrition Agent

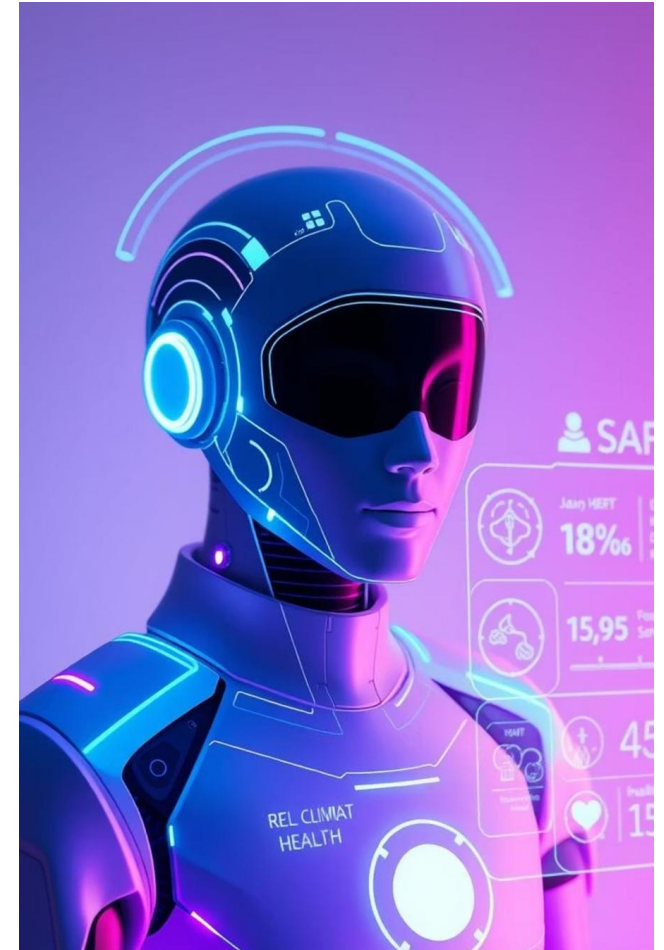
Agentic AI enables the Nutrition Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to deliver personalized nutrition guidance.

Each agent performs a specific role—such as retrieving nutritional data, generating diet plans, providing health advice, and analyzing food logs—while working collaboratively. This **multi-agent approach** ensures efficient task execution and accurate results.

Agentic AI also brings **context awareness**, understanding user details like age, health conditions, and fitness goals to generate tailored recommendations. It supports **real-time adaptation**, continuously updating suggestions based on user inputs like daily meals.

Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **smart digital nutritionist** that helps users maintain a healthy lifestyle.



Novelty and Uniqueness

The proposed **Smart Study Generator Agent** stands out by combining **Agentic AI, RAG, and personalized learning** into a single intelligent study assistant.

- **Multi-Agent Intelligence** – Multiple specialized agents collaborate to generate study plans, summarize concepts, create quizzes, and answer subject-specific queries.
- **RAG-Based Accuracy** – Retrieves information from trusted educational resources and combines it with AI-generated responses for accurate and up-to-date learning content.
- **Personalized Learning Paths** – Creates customized study schedules and learning materials based on the student's goals, skill level, and available study time.
- **Adaptive Study Support** – Continuously refines recommendations using student progress, quiz performance, and learning behavior.
- **Multi-Modal Learning** – Supports text, PDFs, images, and voice-based interactions, making learning more engaging and accessible.
- **Interactive Assessment** – Generates practice questions, flashcards, and instant feedback to improve knowledge retention and exam readiness.
- **Scalable & Enterprise-Ready** – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring secure, reliable, and governed AI-powered educational assistance.

Git Hub Link

GitHub Repository Link of Smart Study Generator Agent:- [Link](#)

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Future Scope

- **AI Tutor Integration** – Offer personalized tutoring with step-by-step explanations and instant doubt resolution.
- **Adaptive Learning** – Continuously refine study plans based on student progress, performance, and learning pace.
- **Multi-Language Support** – Provide study content and explanations in multiple regional and global languages.
- **LMS Integration** – Seamlessly integrate with platforms like Google Classroom, Moodle, and Microsoft Teams.
- **Gamification & Assessments** – Introduce quizzes, flashcards, badges, and rewards to increase engagement and knowledge retention.
- **Performance Analytics** – Track learning trends, identify weak areas, and recommend targeted improvement strategies.
- **Multi-Modal Learning** – Support text, voice, image, and PDF-based learning for a more interactive experience.

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Thank You!

Thank you for your time and interest.